PREMIUM

Full-Length Mock Test

Mock 03

Mathematics

GATE MA

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20% | Moderate 33% | Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	55	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		**60**		**100**	

Marking Scheme

- Correct MCQ: +1 mark (or +2 marks as specified per question)
- Wrong MCQ: -0.33 marks
- Correct MSQ: +2 marks
- Wrong MSQ (partial or no match): -0.33 marks
- Correct NAT: +1 mark (or +2 marks as specified per question)
- No answer / Unattempted: 0 marks
- NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Abstract Algebra — Rings]

Kernel of group homomorphism is:

A) Subgroup
B) Normal subgroup
C) Quotient
D) Coset

Q.2 [1 Mark] [EASY] [Real Analysis — Series]

A continuous function on closed bounded set is:

A) Unbounded
B) Bounded
C) Nowhere
D) Discontinuous

Q.3 [1 Mark] [EASY] [Linear Algebra — Eigenvalues]

Rank of identity matrix I_n:

A) 0
B) 1
C) n
D) n^2

Q.4 [1 Mark] [EASY] [Topology — Continuity]

Kernel of group homomorphism is:

A) Subgroup
B) Normal subgroup
C) Quotient
D) Coset

Q.5 [1 Mark] [EASY] [ODE & PDE — Second Order ODE]

A continuous function on closed bounded set is:

A) Unbounded
B) Bounded
C) Nowhere
D) Discontinuous

Q.6 [1 Mark] [EASY] [Abstract Algebra — Lagrange Theorem]

Rank of identity matrix I_n:

A) 0
B) 1
C) n
D) n^2

Q.7 [1 Mark] [EASY] [Real Analysis — Riemann Integral]

Kernel of group homomorphism is:

A) Subgroup

B) Normal subgroup

C) Quotient

D) Coset

Q.8 [1 Mark] [EASY] [Linear Algebra — Diagonalization]

A continuous function on closed bounded set is:

A) Unbounded
B) Bounded
C) Nowhere
D) Discontinuous

Q.9 [1 Mark] [EASY] [Topology — Continuity]

Rank of identity matrix I_n:

A) 0
B) 1
C) n
D) n^2

Q.10 [1 Mark] [EASY] [ODE & PDE — Wave Equation]

Kernel of group homomorphism is:

A) Subgroup
B) Normal subgroup
C) Quotient
D) Coset

Q.11 [1 Mark] [EASY] [Abstract Algebra — Rings]

A continuous function on closed bounded set is:

A) Unbounded
B) Bounded
C) Nowhere
D) Discontinuous

Q.12 [1 Mark] [EASY] [Real Analysis — Series]

Rank of identity matrix I_n:

A) 0
B) 1
C) n
D) n^2

Q.13 [2 Marks] [MODERATE] [Linear Algebra — Inner Products]

Kernel of group homomorphism is:

A) Subgroup
B) Normal subgroup
C) Quotient
D) Coset

Q.14 [2 Marks] [MODERATE] [Topology — Metric Spaces]

A continuous function on closed bounded set is:

A) Unbounded

B) Bounded

C) Nowhere

D) Discontinuous

Q.15 [2 Marks] [MODERATE] [ODE & PDE — Wave Equation]

Rank of identity matrix I_n:

A) 0
B) 1
C) n
D) n^2

Q.16 [2 Marks] [MODERATE] [Abstract Algebra — Fields]

Kernel of group homomorphism is:

A) Subgroup
B) Normal subgroup
C) Quotient
D) Coset

Q.17 [2 Marks] [MODERATE] [Real Analysis — Series]

A continuous function on closed bounded set is:

A) Unbounded
B) Bounded
C) Nowhere
D) Discontinuous

Q.18 [2 Marks] [MODERATE] [Linear Algebra — Linear Maps]

Rank of identity matrix I_n:

A) 0
B) 1
C) n
D) n^2

Q.19 [2 Marks] [MODERATE] [Topology — Metric Spaces]

Kernel of group homomorphism is:

A) Subgroup
B) Normal subgroup
C) Quotient
D) Coset

Q.20 [2 Marks] [MODERATE] [ODE & PDE — Second Order ODE]

A continuous function on closed bounded set is:

A) Unbounded
B) Bounded
C) Nowhere
D) Discontinuous

Q.21 [2 Marks] [MODERATE] [Abstract Algebra — Rings]

Rank of identity matrix I_n:

A) 0

B) 1

C) n

D) n^2

Q.22 [2 Marks] [MODERATE] [Real Analysis — Continuity]

Kernel of group homomorphism is:

A) Subgroup
B) Normal subgroup
C) Quotient
D) Coset

Q.23 [2 Marks] [MODERATE] [Linear Algebra — Eigenvalues]

A continuous function on closed bounded set is:

A) Unbounded
B) Bounded
C) Nowhere
D) Discontinuous

Q.24 [2 Marks] [MODERATE] [Topology — Connectedness]

Rank of identity matrix I_n:

A) 0
B) 1
C) n
D) n^2

Q.25 [2 Marks] [MODERATE] [ODE & PDE — Laplace Transforms]

Kernel of group homomorphism is:

A) Subgroup
B) Normal subgroup
C) Quotient
D) Coset

Q.26 [2 Marks] [MODERATE] [Abstract Algebra — Groups]

A continuous function on closed bounded set is:

A) Unbounded
B) Bounded
C) Nowhere
D) Discontinuous

Q.27 [2 Marks] [MODERATE] [Real Analysis — Riemann Integral]

Rank of identity matrix I_n:

A) 0
B) 1
C) n
D) n^2

Q.28 [2 Marks] [MODERATE] [Linear Algebra — Vector Spaces]

Kernel of group homomorphism is:

A) Subgroup

B) Normal subgroup

C) Quotient

D) Coset

Q.29 [2 Marks] [MODERATE] [Topology — Metric Spaces]

A continuous function on closed bounded set is:

A) Unbounded
B) Bounded
C) Nowhere
D) Discontinuous

Q.30 [2 Marks] [MODERATE] [ODE & PDE — First Order ODE]

Rank of identity matrix I_n:

A) 0
B) 1
C) n
D) n^2

Q.31 [2 Marks] [MODERATE] [Abstract Algebra — Fields]

Kernel of group homomorphism is:

A) Subgroup
B) Normal subgroup
C) Quotient
D) Coset

Q.32 [2 Marks] [MODERATE] [Real Analysis — Sequences]

A continuous function on closed bounded set is:

A) Unbounded
B) Bounded
C) Nowhere
D) Discontinuous

Q.33 [2 Marks] [HARD] [Linear Algebra — Eigenvalues]

Rank of identity matrix I_n:

A) 0
B) 1
C) n
D) n^2

Q.34 [2 Marks] [HARD] [Topology — Continuity]

Kernel of group homomorphism is:

A) Subgroup
B) Normal subgroup
C) Quotient
D) Coset

Q.35 [2 Marks] [HARD] [ODE & PDE — Wave Equation]

A continuous function on closed bounded set is:

A) Unbounded

B) Bounded

C) Nowhere

D) Discontinuous

Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Abstract Algebra — Fields]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.37 [2 Marks] [HARD] [Real Analysis — Differentiability]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.38 [2 Marks] [HARD] [Linear Algebra — Inner Products]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.39 [2 Marks] [HARD] [Topology — Metric Spaces]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.40 [2 Marks] [HARD] [ODE & PDE — Wave Equation]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Section C — Numerical Answer Type (NAT)

Q.51 [2 Marks] [HARD] [Abstract Algebra — Rings]

Standard calculation using basic formula:

Answer: ____________________

Q.52 [2 Marks] [HARD] [Real Analysis — Riemann Integral]

Standard calculation using basic formula:

Answer: ____________________

Q.53 [2 Marks] [HARD] [Linear Algebra — Diagonalization]

Standard calculation using basic formula:

Answer: ____________________

Q.54 [2 Marks] [HARD] [Topology — Metric Spaces]

Standard calculation using basic formula:

Answer: ____________________

Q.55 [2 Marks] [HARD] [ODE & PDE — Laplace Transforms]

Standard calculation using basic formula:

Answer: ____________________

Q.56 [2 Marks] [HARD] [Abstract Algebra — Homomorphisms]

Standard calculation using basic formula:

Answer: ____________________

Q.57 [2 Marks] [HARD] [Real Analysis — Continuity]

Standard calculation using basic formula:

Answer: ____________________

Q.58 [2 Marks] [HARD] [Linear Algebra — Diagonalization]

Standard calculation using basic formula:

Answer: ________________________

Q.59 [2 Marks] [HARD] [Topology — Continuity]

Standard calculation using basic formula:

Answer: ____________________

Q.60 [2 Marks] [HARD] [ODE & PDE — Wave Equation]

Standard calculation using basic formula:

Answer: ____________________

Answer Key

Verify your answers after completing the full paper.

Q.1:	B	**Q.2:**	B	**Q.3:**	C
Q.4:	B	**Q.5:**	B	**Q.6:**	C
Q.7:	B	**Q.8:**	B	**Q.9:**	C
Q.10:	B	**Q.11:**	B	**Q.12:**	C
Q.13:	B	**Q.14:**	B	**Q.15:**	C
Q.16:	B	**Q.17:**	B	**Q.18:**	C
Q.19:	B	**Q.20:**	B	**Q.21:**	C
Q.22:	B	**Q.23:**	B	**Q.24:**	C
Q.25:	B	**Q.26:**	B	**Q.27:**	C
Q.28:	B	**Q.29:**	B	**Q.30:**	C
Q.31:	B	**Q.32:**	B	**Q.33:**	C
Q.34:	B	**Q.35:**	B	**Q.36:**	AC
Q.37:	AC	**Q.38:**	AC	**Q.39:**	AC
Q.40:	AC	**Q.41:**	AC	**Q.42:**	AC
Q.43:	AC	**Q.44:**	AC	**Q.45:**	AC
Q.46:	AC	**Q.47:**	AC	**Q.48:**	AC
Q.49:	AC	**Q.50:**	AC	**Q.51:**	10
Q.52:	10	**Q.53:**	10	**Q.54:**	10
Q.55:	10	**Q.56:**	10	**Q.57:**	10
Q.58:	10	**Q.59:**	10	**Q.60:**	10

Detailed Solutions

Question 1
[Abstract Algebra — Rings]

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 2
[Real Analysis — Series]

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set !' attains max/min.

Question 3
[Linear Algebra — Eigenvalues]

Rank of identity matrix I_n:

Answer: (C)

Identity matrix I_n has rank n.

Question 4
[Topology — Continuity]

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 5
[ODE & PDE — Second Order ODE]

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set !' attains max/min.

Question 6
[Abstract Algebra — Lagrange Theorem]

Rank of identity matrix I_n:

Answer: (C)

Identity matrix I_n has rank n.

Question 7
[Real Analysis — Riemann Integral]

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 8
[Linear Algebra — Diagonalization]

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set !' attains max/min.

Question 9
[Topology — Continuity]

Rank of identity matrix I_n:

Answer:

(C)

Identity matrix I_n has rank n.

Question 10 *[ODE & PDE — Wave Equation]*

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 11 *[Abstract Algebra — Rings]*

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set !' attains max/min.

Question 12 *[Real Analysis — Series]*

Rank of identity matrix I_n:

Answer: (C)

Identity matrix I_n has rank n.

Question 13 *[Linear Algebra — Inner Products]*

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 14 *[Topology — Metric Spaces]*

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set !' attains max/min.

Question 15 *[ODE & PDE — Wave Equation]*

Rank of identity matrix I_n:

Answer: (C)

Identity matrix I_n has rank n.

Question 16 *[Abstract Algebra — Fields]*

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 17 *[Real Analysis — Series]*

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set !' attains max/min.

Question 18 *[Linear Algebra — Linear Maps]*

Rank of identity matrix I_n:

Answer:

(C)

Identity matrix I_n has rank n.

Question 19 *[Topology — Metric Spaces]*

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 20 *[ODE & PDE — Second Order ODE]*

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set !' attains max/min.

Question 21 *[Abstract Algebra — Rings]*

Rank of identity matrix I_n:

Answer: (C)

Identity matrix I_n has rank n.

Question 22 *[Real Analysis — Continuity]*

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 23 *[Linear Algebra — Eigenvalues]*

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set !' attains max/min.

Question 24 *[Topology — Connectedness]*

Rank of identity matrix I_n:

Answer: (C)

Identity matrix I_n has rank n.

Question 25 *[ODE & PDE — Laplace Transforms]*

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 26 *[Abstract Algebra — Groups]*

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set !' attains max/min.

Question 27 *[Real Analysis — Riemann Integral]*

Rank of identity matrix I_n:

Answer:

(C)

Identity matrix I_n has rank n.

Question 28 *[Linear Algebra — Vector Spaces]*

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 29 *[Topology — Metric Spaces]*

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set !' attains max/min.

Question 30 *[ODE & PDE — First Order ODE]*

Rank of identity matrix I_n:

Answer: (C)

Identity matrix I_n has rank n.

Question 31 *[Abstract Algebra — Fields]*

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 32 *[Real Analysis — Sequences]*

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set !' attains max/min.

Question 33 *[Linear Algebra — Eigenvalues]*

Rank of identity matrix I_n:

Answer: (C)

Identity matrix I_n has rank n.

Question 34 *[Topology — Continuity]*

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 35 *[ODE & PDE — Wave Equation]*

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set !' attains max/min.

Question 36 *[Abstract Algebra — Fields]*

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 37
[Real Analysis — Differentiability]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 38
[Linear Algebra — Inner Products]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 39
[Topology — Metric Spaces]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 40
[ODE & PDE — Wave Equation]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 41
[Abstract Algebra — Lagrange Theorem]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 42
[Real Analysis — Riemann Integral]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 43
[Linear Algebra — Inner Products]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 44
[Topology — Connectedness]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 45
[ODE & PDE — Second Order ODE]

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 46 *[Abstract Algebra — Fields]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 47 *[Real Analysis — Series]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 48 *[Linear Algebra — Vector Spaces]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 49 *[Topology — Compactness]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 50 *[ODE & PDE — Second Order ODE]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 51 *[Abstract Algebra — Rings]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 52 *[Real Analysis — Riemann Integral]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 53 *[Linear Algebra — Diagonalization]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 54 *[Topology — Metric Spaces]*

Standard calculation using basic formula:

Answer:

(10)

Apply the appropriate formula for the given data.

Question 55 *[ODE & PDE — Laplace Transforms]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 56 *[Abstract Algebra — Homomorphisms]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 57 *[Real Analysis — Continuity]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 58 *[Linear Algebra — Diagonalization]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 59 *[Topology — Continuity]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 60 *[ODE & PDE — Wave Equation]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.
